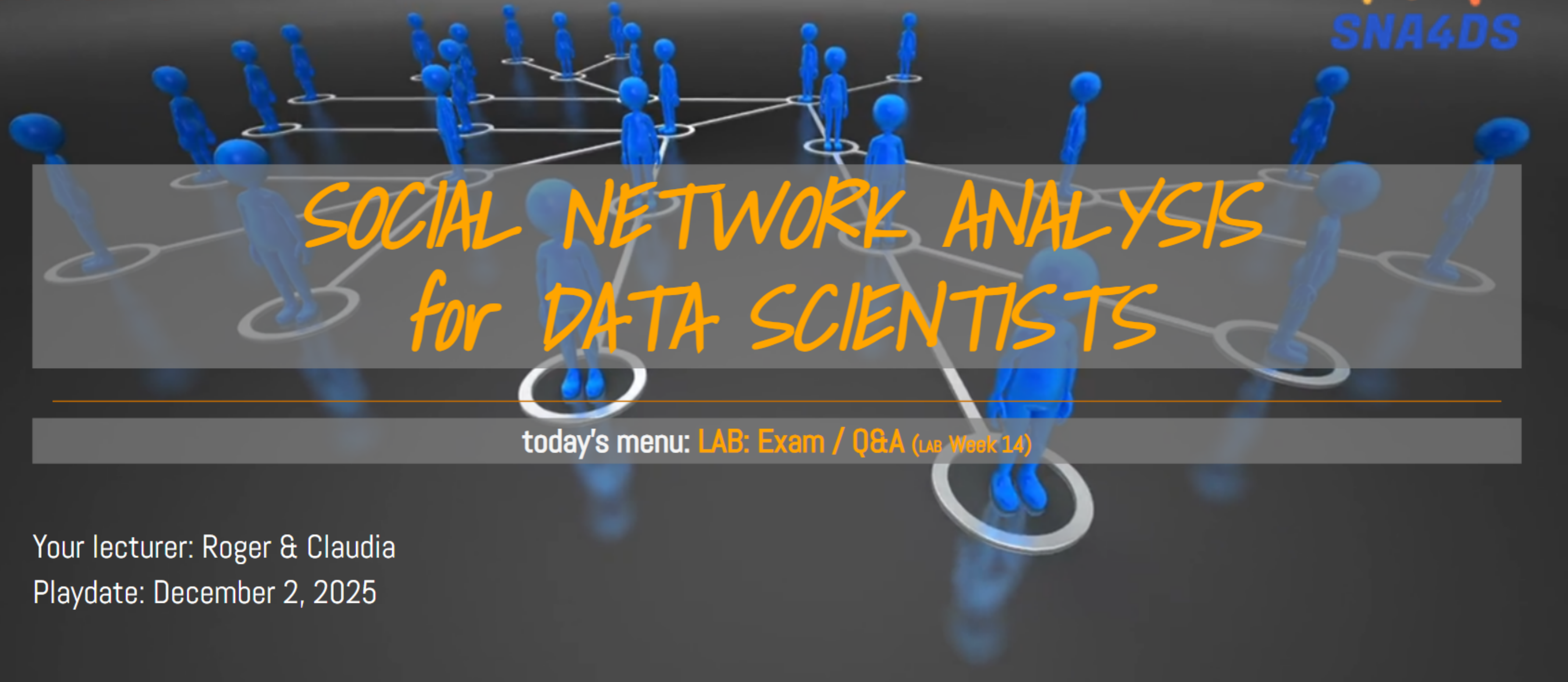




SOCIAL NETWORK ANALYSIS for DATA SCIENTISTS

today's menu: LAB: Exam / Q&A (LAB Week 14)

Your lecturer: Roger & Claudia

Playdate: December 2, 2025

ID: 50197 4 punten

You want to analyze the support politicians give each other in parliament. Your dependent measure is the number of times one politician voted in support of the other's proposal/bill over the past three months and you construct a matrix with these numbers (the diagonal, of course, is not defined here).

Your first hypothesis is that politicians who are a member of the same party ("same/different party") will support each other's proposals/bills. But you also believe that friendship strength plays a role (measured on a 7-point scale) and the number of times the politician supported the other politician's proposals over the previous two years (numeric).

You are considering which of the following models to use:

- OLS
- QAP test
- QAP linear model
- QAP logistic model
- network autocorrelation
- Conditional Uniform Graph

Explain which the most appropriate statistical method for this purpose is and explain why.

Antwoord intypen

Geen antwoord ingeleverd

You want to analyze the support politicians give each other in parliament. Your dependent measure is whether a politician voted in favor ("yes/no") of the other's proposal/bill at least once in the past three months and you construct a matrix with this scores (the diagonal, of course, is not defined here).

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You want to analyze the support politicians give each other in parliament. In particular, you want to see to what extent politicians "clique together": if John supports Peter, and Peter supports Mary, then John tends to support Mary too. This is a structure you would expect in parliaments where groups of politicians support each other and trade their support of the other in favor of the other supporting them in return and where a "friend-of-a friend-is-my-friend" (and, vice versa, of course, because this is still pure politics and deal-making).

You have data on whether one politician voted in favor ("yes/no") of the other's proposal/bill at least once in the past three months and you construct a matrix with this scores (the diagonal, of course, is not defined here). Your hypothesis is that this mutual-support mechanism occurs more than you would expect by chance (supporting the idea that politicians behave differently than "regular people")

You are considering which of the following models to use:

- OLS
- QAP test
- QAP linear model
- QAP logistic model
- network autocorrelation
- Conditional Uniform Graph

Explain which measures you use to test your hypothesis and what is the most appropriate statistical method for this purpose is and explain why.

ID: 50206 6 punten

This question has two subquestion:

1. Explain how SNA is useful in the academic field of "entrepreneurship."
2. Formulate an academically relevant network research question in this field and briefly explain how you can research this using the tools from this course.

Antwoord intypen

Geen antwoord ingeleverd



You have the following network:

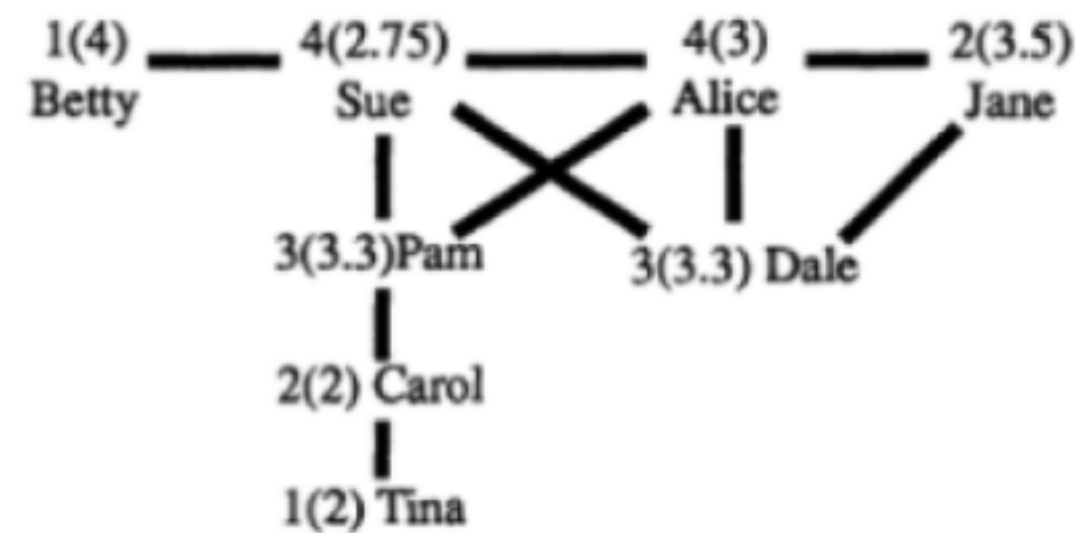




People expect themselves and others to have friends and often wonder about the normality of those individuals who appear to have few or no friends. Research on how many friends people consider to be an "adequate" number, it is reasonable to suppose that individuals compare the number of friends that their friends have a relevant basis of comparison. In other words: if I have about the same number of friends that my friends do, I have an adequate number of friends.

However, the reality is that if individuals indeed make this type of comparison, it is likely that most of them will feel relatively inadequate. A well-known phenomenon from friendship networks is that the majority of people experience that "my friends have, on average, more friends than I do". This is counterintuitive, because how could it be possible that most people have fewer friends than their friends average?

Below you see a network of eight girls, with an edge representing a friendship. The number beside/above each name is the number of friends from the girl. The number in parentheses is the mean number of friends of her friends. So, Jane has 2 friends and these two friends average $(3 + 4)/2 = 3.5$ friends.



Here,

- Betty's only friend, Sue, has more friends than Betty has
- Jane's two friends, Dale and Alice, average more friends than Jane has
- Dale's three friends, Sue, Alice, and Jane, average more friends than Dale

and so forth. Of the eight girls, five (Betty, Jane, Pam, Dale, and Tina) have fewer friends than the average among their friends, while only two (Sue and Alice) have more friends than the average among their friends; one (Carol) has as many as the average among her friends.

In sum, most of these girls have fewer friends than their friends average. This is a phenomenon well-known in friendship networks, this example is quite typical of those.

While it is not a mathematical necessity that each individual will have fewer friends than the mean of her or his own friends, most people will find themselves in this situation.

The basic logic can be described simply.

If there are some people with many friendship ties and others with few (which is to be expected in most friendship networks), those with many ties show up disproportionately in sets of friends. For example, those with 40 friends show up in each of 40 individual friendship networks and thus can make 40 people feel relatively deprived, while those with only one friend show up in only one friendship network and can make only that one person feel relatively advantaged. Thus, it is inevitable that individual friendship networks disproportionately include those with the most friends.

In lingo: this is due to the skewness of the degree distribution in this undirected graph.

You work for a local government, in a job where you are responsible for the sewage system. In the Corona times, budgets are limited, since the government spends a considerable part of its budget to support businesses that had to close as part of the lockdowns in 2020.

Given the limited budget, you want to identify the pipes that are most likely to experience cracks or deterioration, due to the flow of the sewage water (which has also increased during the Corona epidemic). Unfortunately, you have no sensors in the sewage pipes that allow you to automatically keep track of the state of each pipe.

Rather, you realize that SNA can help you identify which pipes experience the highest strain as a result of the sewage water flow. For simplicity, assume that this is the only relevant variable for now (in a actual analysis, you would also include the material the pipe is made of, its age, its angle, etc.).

Which centrality measure (discussed in the course) finds you the pipe that is likely to be most affected by sewage water flow (with respect to this single objective)? Carefully explain why.

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ataset: conference

The dataset is network data of interactions of 75 academists at a scientific conference. The actual temporal network dataset is called "conference_nw". Data were automatically collected through RFID tags (ie., wearable badges that detect the presence of other badges within 0.5 metre distance). All interactions were measured per 1/10 of a second interval (so, time = 10 means that 1 second has passed).

A [3p] The measurement started at 8.30am. At what time did the last interaction end? Round to the nearest quarter of an hour.

B [3p] Which vertex represents Johnny Le (coded as "Le, Johnny")?

C [6p] At 10:00am coffee was served in a separate meeting room. About half of the conference participants go there, including Johnny Le. How many people does Johnny Le interact with between 10:00am and 10:30am? How many interactions occur overall within this interval between 10:00am and 10:30am? If everybody who goes to the coffee serving interacts with at least one other person, how many conference participants went to the coffee room?

D [8p] Make a plot of the evolution of density of the conference network over the entire observation period. For this purpose, collapse the network into slices of 15 minutes, at 5 minute intervals (so, you plot the density of intervals 0-15, 5-20, 10-25, et cetera., all the way until the end)



```
A
max(as.data.frame(conference_nw)$terminus)/10/60/60 # 9 hours and 15 mins -> 17:45h

B # then see where Johnny is: 31
SNA4DS::extract_all_vertex_attributes(networkDynamic::network.collapse(conference_nw))

C
# How many interactions occur overall within this interval?
start <- 90*60*10
end <- 120*60*10
coffee <- networkDynamic::network.collapse(conference_nw, onset = start, terminus = end)
coffee # you can see that there were 326 edges from the output
coffee_socio <- network::as.sociomatrix(coffee)
sum(coffee_socio) # also 326

# How many people does Johnny interact with between 1000 and 1030?
rowSums(coffee_socio)[31] # he interacts with 11 people

# If everybody who goes there interacts with at least one other person, how many
# conference participants went to the coffee room?
sum(rowSums(coffee_socio) > 0) # 28 people
# or via
table(rowSums(coffee_socio))

D
plot(tsna::tSnaStats(conference_nw, snafun = "gden", time.interval = 5*60*10, aggregate.dur = 15*60*10))
```